DEEP DIVE

When Drones Hit the Cloud

The AWS Middle East strikes and the end of the safe infrastructure assumption

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FOREWORD

On March 1, Iranian drone strikes hit AWS data centres in the UAE and Bahrain — taking 109 services offline, shutting down banking and payment platforms across the Gulf, and introducing a category of infrastructure risk that no resilience playbook was designed to handle. The cloud's longest-standing thought experiment became operational reality: what happens when someone physically destroys the data centre your business runs on?

We're dedicating a full issue to this because the implications extend far beyond a service outage. This event changes how engineering leaders think about geographic risk, disaster recovery assumptions, and whether the legal frameworks governing data residency can survive contact with the physical world. The CSIS called it a paradigm shift. They're not wrong.

How to Read This Issue

2 min	Read the executive summary and the key takeaway box at the end
10 min	Add the foreword, cascade diagram, architecture visual, and lessons section
30 min	Read the full issue end to end, including industry response and sources

AWS ME RECOVERY ~40% services restored · Week 3 · No full recovery date

EXECUTIVE SUMMARY

What Happened and Why It Matters

On March 1, 2026, Iranian retaliatory drone strikes hit two Amazon Web Services data centres in the UAE and damaged a third facility in Bahrain. Two of three availability zones in the ME-CENTRAL-1 region went offline — not degraded, offline. Over 109 services were disrupted. Banking apps, payment platforms, and enterprise software across the Gulf went dark. Recovery has taken weeks, not hours — and as of March 18, full restoration has not been confirmed.

This was the first time a major hyperscale cloud provider's infrastructure was directly hit by military action. It introduced a risk category the industry had theorised about but never experienced: kinetic attacks on the digital infrastructure that underpins the Gulf's financial system. AWS initially described the cause as "objects" before confirming drone strikes 36 hours later — then issued an advisory unprecedented in the company's history, telling customers to migrate workloads to alternate regions entirely.

For engineering teams operating in the region — particularly those in banking, payments, and fintech — the implications are structural. Multi-AZ redundancy within a single region, the standard resilience playbook, assumed that availability zones fail independently through technical faults. That assumption broke when two of three zones were hit simultaneously by coordinated strikes, with a third zone in a neighbouring country also damaged the same day.

The data residency paradox made it worse. UAE and Bahrain central bank mandates require financial data to stay in-region. But the region's infrastructure was physically destroyed. Failing over to another region resolves the technical outage but creates a legal one. Many organisations discovered at 4:30 AM that they had no pre-approved authority for cross-border failover under emergency conditions.

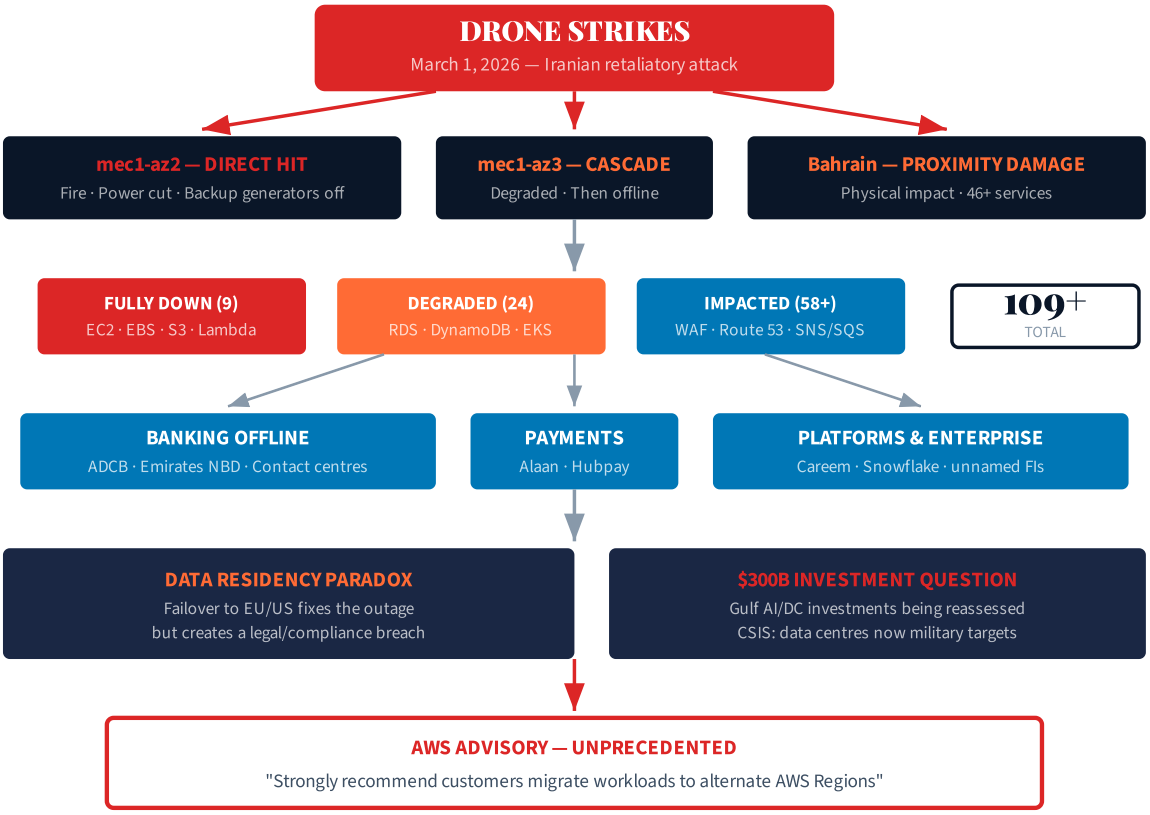
109+	3	2/3	$300B
SERVICES DISRUPTED	FACILITIES HIT	UAE AZS OFFLINE	GULF DC INVESTMENT AT RISK

CASCADE IMPACT

How the Strikes Rippled Through the Stack

The damage didn't stop at the data centres. A single kinetic event cascaded through availability zones, cloud services, downstream platforms, financial infrastructure, and legal frameworks — each

layer amplifying the impact of the one before it.



How the March 1 drone strikes cascaded from physical infrastructure through 109+ cloud services to banking, payments, legal, and investment consequences

THE FULL TIMELINE

From First Strike to Week Three

PHASE 1 — MARCH 1: THE STRIKES

4:30 AM PST — "Objects" strike the mec1-az2 availability zone in UAE. Fire breaks out inside the data centre. Local fire department responds and shuts off all power — including backup generators — to contain the blaze.

6:09 AM — AWS confirms power failure in mec1-az2. EC2 instances, EBS volumes, and RDS databases in the zone become unavailable. AWS frames it as a "localized power issue."

11:30 AM — Cascading failure. The second availability zone (mec1-az3) enters a degraded state. Only mec1-az1 remains partially operational — but EC2 API errors and launch failures are reported there too. Two of three UAE AZs are now offline.

9:56 PM — ME-SOUTH-1 (Bahrain) hit. A drone strike in close proximity to an AWS facility causes physical damage. 46+ services affected in Bahrain. The attack surface now spans

two countries.

PHASE 2 — MARCH 2–3: CONFIRMATION & ADVISORY

March 2, 7:19 PM EST — AWS breaks its silence on the cause: "In the UAE, two of our facilities were directly struck, while in Bahrain, a drone strike in close proximity to one of our facilities caused physical impacts to our infrastructure. These strikes have caused structural damage, disrupted power delivery, and in some cases required fire suppression activities that resulted in additional water damage."

March 3, 8:14 AM — Partial S3 recovery begins. New PUT and LIST operations work; GET operations for pre-existing data remain dependent on physical restoration. EC2 instance launches throttled. DynamoDB error rates elevated.

March 3 — AWS issues its strongest-ever advisory: "We continue to strongly recommend that customers with workloads running in the Middle East take action now to migrate those workloads to alternate AWS Regions."

PHASE 3 — MARCH 4 ONWARDS: PROLONGED RECOVERY

Week of March 3–10 — Physical repair of cooling systems, power infrastructure, and structural damage continues. AWS pursues parallel recovery: physical restoration alongside software mitigations to restore partial availability.

Week of March 10–17 — AWS re:Post forum shows customers still unable to retrieve snapshots or migrate workloads from affected zones. Snowflake reports regional outages affecting enterprise data. No confirmed full restoration date.

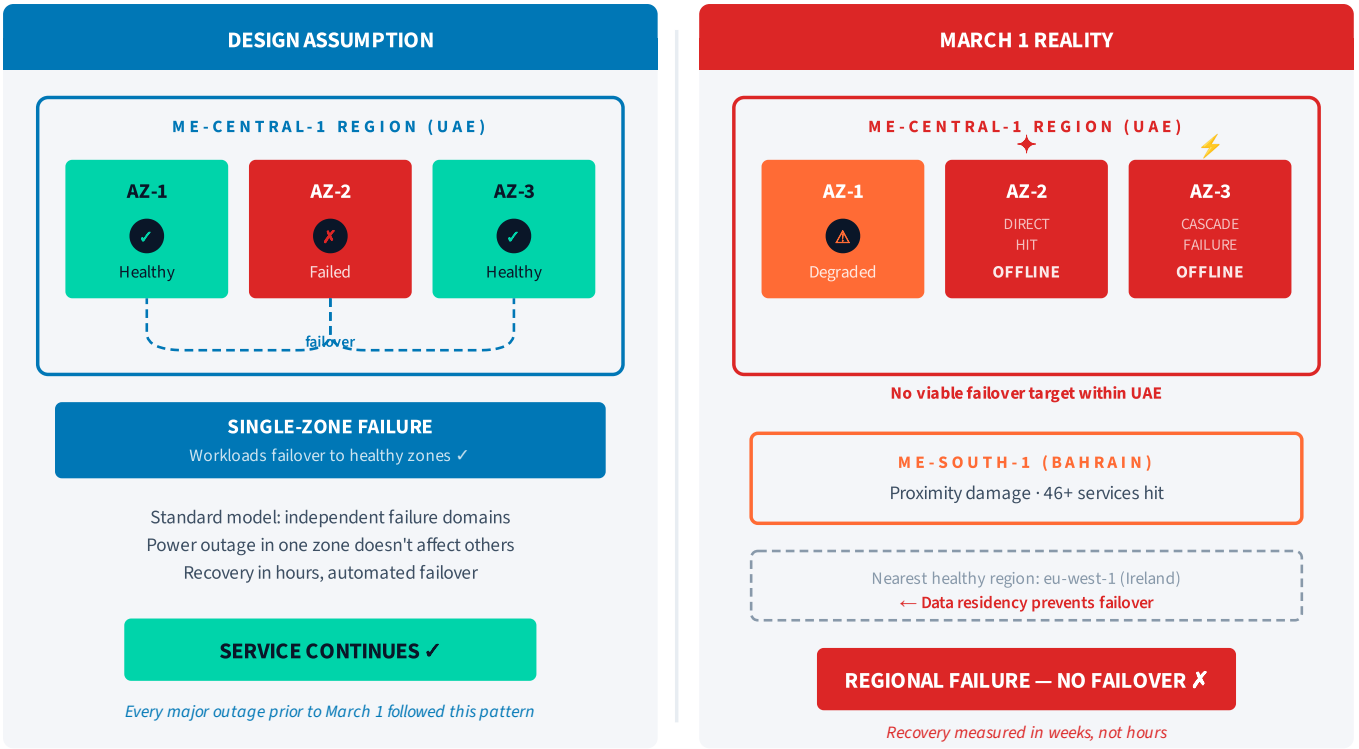
March 18 (today) — Approximately 40% of services restored. Full recovery timeline unknown. AWS continues to recommend alternate regions.

ARCHITECTURE ANALYSIS

The Multi-AZ Assumption vs. Reality

AWS's multi-AZ architecture is designed to survive the failure of a single availability zone — the standard resilience model assumes independent failure domains. On March 1, that assumption was

invalidated when drone strikes hit two of three UAE zones simultaneously and damaged a third in a neighbouring country.



Standard multi-AZ resilience model (left) assumes single-zone failure with automatic failover. March 1 reality (right): two zones hit simultaneously, third degraded, neighbouring region also damaged — no viable failover within data residency boundaries.

> *"In previous conflicts, adversaries targeted pipelines, refineries, and oil fields. In the compute era, these actors could also target data centres, energy infrastructure supporting compute, and fibre chokepoints."*
>
> — Center for Strategic and International Studies (CSIS), March 2026

IMPACT ANALYSIS

Who Was Affected and How

Service tiers at peak disruption

Fully disrupted (9 services): EC2 (networking APIs returning 500s), EBS (volumes inaccessible), S3 (high failure rates), Lambda (invocations failing), Management Console (partial access only).

Degraded (24+ services): RDS, DynamoDB, Cognito (authentication), EKS (cluster management), VPC networking.

Impacted (58+ services): CloudFormation, Route 53, CloudWatch, Kinesis, WAF, SNS/SQS, and dozens more — cascading failures from the core compute and storage layer.

Banking and payments

The financial services impact was immediate and widespread. ADCB mobile banking and contact centre went offline. Emirates NBD phone banking was disrupted. Payment platforms Alaan and Hubpay went down entirely — not degraded, unavailable. Careem, the region's dominant ride-hailing and delivery platform, was disrupted. Snowflake reported regional outages affecting enterprise data and analytics workloads. Reuters reported multiple unnamed financial institutions were affected.

For banking engineering teams, the outage exposed a specific vulnerability: many organisations in the UAE had chosen single-region deployments to meet data residency requirements under TDRA guidelines. The architecture that satisfied the regulator is the same architecture that had no failover path when the region went down.

> **WHAT IS MULTI-AZ REDUNDANCY?**
>
> AWS organises each geographic region into multiple "availability zones" — physically separate data centres within the same metro area, connected by low-latency networking. The standard resilience pattern is to deploy across at least two AZs so that if one fails through a technical fault, the other continues operating. Multi-AZ is designed for independent, uncorrelated failures: a power outage in one facility, a cooling failure in another. It was not designed for coordinated kinetic attacks affecting an entire geographic area simultaneously.

The data residency paradox

The technical imperative

When two of three availability zones went offline and the region's infrastructure sustained physical damage, the standard engineering response is failover: route traffic to eu-west-1, us-east-1, or any healthy region. Technically, this is straightforward for well-architected applications.

AWS's own advisory made the recommendation explicit: migrate to alternate regions. From a pure availability standpoint, this is correct. The infrastructure was physically damaged. Recovery would take weeks. Customers needed their services running.

The legal problem

UAE and Bahrain central bank mandates require financial data to remain in-region. TDRA compliance frameworks specify data residency. Failing over resolves the outage but potentially violates these requirements — creating a legal exposure at the exact moment you're solving an operational crisis.

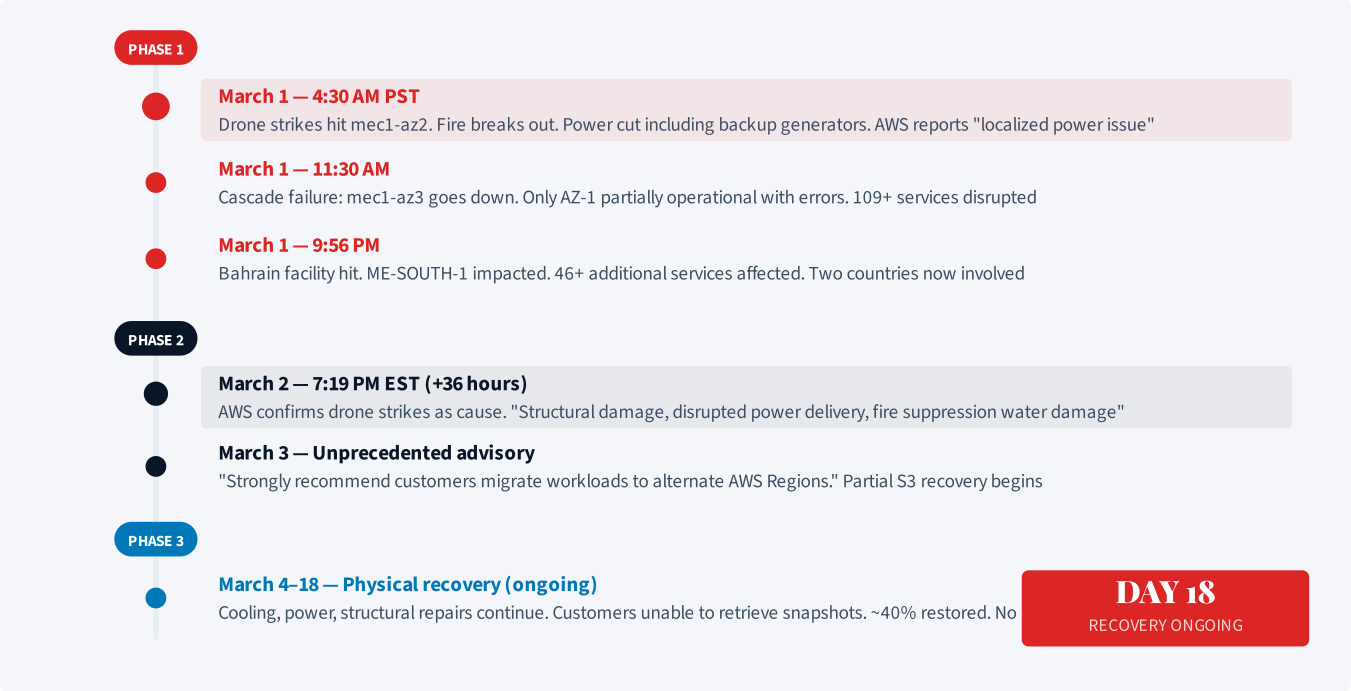
The Dubai Financial Services Authority issued emergency guidance allowing temporary cross-border processing. But many organisations discovered at 4:30 AM that they had no pre-approved authority for emergency data movement. That decision framework needs to exist before the incident.

Sources: CNBC, The Register, Cybersecurity News, Cybele Software

RECOVERY TIMELINE

Three Weeks and Counting

Unlike software outages where recovery is measured in hours, physical damage requires on-site assessment, component sourcing, safety clearances, and structural repair. The March 1 strikes introduced a recovery timeline closer to a construction project than an incident response runbook.



Three phases: strikes and cascade (red), confirmation and advisory (navy), prolonged physical recovery (teal). Day 18 — no confirmed full restoration date.

INDUSTRY RESPONSE

How the Industry Reacted

AWS: 36 hours of ambiguity

AWS initially described the cause as "objects" and a "localized power issue." The company confirmed drone strikes 36 hours later. The gap matters: for a day and a half, customers making decisions about failover, data migration, and incident response were operating without accurate information about the cause, scope, or likely recovery timeline. Whether this was a power outage (hours) or physical destruction (weeks) changes every downstream decision an engineering leader needs to make.

> *We continue to strongly recommend that customers with workloads running in the Middle East take action now to migrate those workloads to alternate AWS Regions. The ongoing conflict in the region means that the broader operating environment in the Middle East remains unpredictable.*
>
> — Amazon Web Services, March 3, 2026

AWS then pursued parallel recovery: physical restoration alongside software mitigations. The statement that "the broader operating environment in the Middle East remains unpredictable" was unprecedented. A hyperscale cloud provider publicly acknowledging that geopolitical instability makes its infrastructure unreliable in a region has no precedent in the industry's history.

Other cloud providers

Google, Microsoft, and Oracle — all of which operate data centres in the region — declined to comment publicly. None reported outages from this event. But the CSIS warning applies equally: roughly 326 data centres across the Middle East sit within the same threat envelope. The physical risk isn't vendor-specific. It's geographic.

The $300B question

Over $300 billion in Gulf AI and data centre investments have been announced in the past 18 months. Those numbers were premised on a stable operating environment. That assumption now has a footnote. For engineering leaders, it translates to a simpler question: will your cloud provider continue to invest in — and maintain — infrastructure in a region where data centres are viable military targets?

Sources: The Register, CNBC, Marketplace, Data Center Knowledge

LESSONS & IMPLICATIONS

What This Changes

A new category of cloud risk

Every previous hyperscale cloud outage — including the US-EAST-1 disruption in October 2025 — was operational in nature: misconfigurations, software bugs, power grid failures. These are well-understood failure modes with well-understood recovery paths. Physical destruction of data centre infrastructure by military action is a fundamentally different category. The recovery timeline isn't measured in incident response playbooks — it's measured in construction schedules. Every disaster recovery plan that assumed "worst case: a few hours" is incomplete.

Multi-AZ is not multi-risk

The standard three-AZ model was designed to survive the failure of any single zone through independent technical faults. It was not designed for two zones being simultaneously struck by coordinated attacks, with the remaining zone degraded from cascading load. For engineering leaders, the distinction matters: multi-AZ protects against operational failures. It does not protect against geographic risks that affect an entire metropolitan area or country.

Vendor diversification doesn't eliminate geographic risk

Spreading workloads across AWS, Azure, and GCP doesn't help if all three operate facilities within the same conflict zone. Multi-cloud in the same geography is diversification theatre. Real geographic resilience means workloads that can run in genuinely separate physical locations — which, for organisations with data residency requirements, circles back to the legal question that has no comfortable answer.

Data centres as strategic targets

CSIS framed this explicitly: in the compute era, adversaries who previously targeted pipelines and refineries will also target data centres, energy infrastructure supporting compute, and fibre chokepoints. This isn't speculative — it's a retrospective observation based on what just happened. For any organisation operating critical infrastructure in a geopolitically contested region, the risk calculus has permanently changed. The question is no longer whether this could happen, but where it happens next.

KEY TAKEAWAYS

→ **Kinetic risk is now a cloud risk category.** Physical destruction of cloud infrastructure by military action has moved from theoretical to demonstrated.

→ **Multi-AZ resilience has a boundary condition.** Coordinated strikes across availability zones bypass the independence assumption that the model depends on.

→ **Multi-cloud in the same geography is not diversification.** The risk is physical proximity to conflict, not vendor dependency.

→ **Data residency and resilience are now in direct tension.** The legal authority for emergency cross-border failover must be established before the incident.

→ **Recovery timelines for physical damage are measured in weeks.** Teams accustomed to hours-long software recovery need fundamentally different planning.

Sources: The Register, CNBC, Data Center Knowledge, Cybersecurity News, AWS re:Post

OTHER NEWS THIS WEEK

In Brief

Atlassian cut 1,600 jobs (10% of workforce) to fund AI pivot — restructuring to "invest aggressively" in AI across Jira, Confluence, and Rovo. Third round of layoffs in two years. *CNBC*

OpenAI released GPT-5.4 — native computer-use capability, 1M token context window. First model to directly control a user's screen. Enterprise pricing at $60/seat/month. *Windows Central*

Google patched two Chrome zero-days (CVE-2026-3909, CVE-2026-3910) — exploited in the wild. V8 type confusion and sandbox escape chain. Patch immediately. *The Hacker News*

Revolut secured full UK banking licence — PRA approval after four-year process. 13M UK customers now FSCS-protected up to £120,000. $75B valuation. Credit products coming. *Reuters, CNBC*

Mistral released Small 4 under Apache 2.0 — open-weight model competitive with GPT-4o at inference. Self-hostable. Relevant for teams with data sovereignty constraints. *Mistral AI*

ON THE RADAR

Coming Up

DEADLINES & REQUIRED ACTIONS

Apr 19	Claude Haiku 3 model retirement — migrate to Haiku 4.5
Apr 30	Node.js 20 end of life — upgrade to Node 22 or 24 LTS
Aug 2	EU AI Act high-risk AI system obligations take effect

EVENTS & CONFERENCES

Apr 22	AWS Summit London — expect Bedrock, Trainium, and ME recovery announcements
Apr 29	Zero Day Con, Dublin — Ireland's largest cybersecurity conference
May 1	Microsoft Copilot Wave 3 / Agent 365 launch

RECOMMENDED READING

Sources & Further Reading

AWS says drones hit two of its datacenters in UAE — The Register

Banking, payments disrupted after AWS UAE drone strikes	CNBC
Amazon cloud unit says drone strikes damaged UAE, Bahrain facilities	Reuters
AWS Middle East Outage After Data Centers Hit by Drone Strikes	Data Center Knowledge
Drone Strikes Highlight Data Center Vulnerabilities	Marketplace
AWS ME-CENTRAL-1: 109 Services Disrupted After Drone Strikes	Cybersecurity News
AWS UAE Drone Strike Outage: DR Lessons Every Cloud Architect Needs	Medium
Service disruption in AWS Middle East Region	AWS re:Post

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